

1. (a) Write the expression for the 'Clapeyron' equation for liquid to vapour transition. (2+2)
(b) Which Maxwell relation according to the lecture notes was used to derive the above 'Clapeyron' Equation?
2. (a) Write the expression for the 'Clausius-Clapeyron' equation for liquid to vapour transition (2+2)
(b) What assumptions/approximations were made to arrive at the above 'Clausius-Clapeyron' equation
(c) Why 'Clausius-Clapeyron' equation connects theory with practical/Laboratory?
3. Write the (a) van der Waals equation of state for a non-ideal gas and what does V stand for? (2+2)
(b) Show the PV isotherm for a real gas undergoing condensation to liquid phase.
4. (a) What is Trouton's rule?
(b) How does Trouton's rule help if the transition temperature is known? (2+2)
5. (a) Why Clausius thought entropy as some function of heat & temperature? (2)
6. Sketch a diagram showing why all the heat absorbed by a gaseous system in a piston is not utilised in doing useful work (2)
7. (a) What is the expression for the work done by the system during expansion in the step/stage 2 of the Carnot cycle?
(b) What is the entropy change during step/stage 1 of the Carnot cycle?
(c) What is the relationship between various forms of heat involved in a Carnot cycle? (2+2)
8. (a) What is the expression for efficiency '%' in Carnot cycle in terms of work, heat and temperature of the reservoir & sink?
(b) Give two interpretations of the '%' expression;
(c) What is Carnot theorem? (2+2)
9. (a) How is Peltier effect connected with the Second Law of Thermodynamics?
(b) What role does Peltier effect play in Biology? (3 (2+1))
10. (a) What chemical is contained in the 'Drinking Bird' working model?
(b) How is 'Drinking Bird' working model connected with the Second Law of Thermodynamics? (3 (1+2))